Validated, yet Wrong: Lessons from Four Metabolomics Validation Audits

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***Abstract*—We audited four of our own machine-learning analyses of small public NMR metabolomics cohorts. Each produced a result we initially treated as validated: a permutation p-value at its floor, a feature-removal (ROAR) curve below random removals, or a high subject-grouped cross-validation score. In all four, the biological reading we attached to the score was unsupported or could not be identified. A responder forecast with permutation $p = 0.001$, from a procedure we have not calibrated, did no better than an unfitted score; coupling between its change-based outcome and its baseline features could explain its predictive performance without any forecastable response. Three compounds of uncertain provenance ranked alongside established metabolites in exploratory ROAR comparisons, which cannot adjudicate provenance. For a diabetes classifier, the data could not show whether its performance came from the disease or from the two collection protocols its classes came from. An "obesity" classifier in fact contrasted patients before and after bariatric surgery. In simulated cohorts with no forecastable response and low measurement reliability, a whole-vector label permutation rejected independence in 26% of datasets at $n = 50$ and in 100% at $n = 500$. We argue that a validation report should name the rival explanations for its headline score and state which test addresses each, and we give six questions that do this for small-n metabolomics.**

***Keywords*—validation, metabolomics, machine learning, permutation test, regression to the mean, feature attribution, confounding, research practice**

I. Introduction

The first result we audit is one we were pleased with. From a single preoperative blood sample, we wanted to forecast which bariatric-surgery patients would be "strong responders" a year later. Our model reached a PR-AUC of 0.683 against a prevalence of 0.437. We repeated the whole cross-validation 1,000 times with the training labels shuffled inside each fold, refitting the model each time, and no shuffled run did as well: $p = 0.001$, the smallest value 1,000 replicates can give. By the checks we knew, the result looked validated.

Then we compared it with a fixed score that involves no training at all, built only from each patient's baseline values and the expected direction of each marker. That score reached 0.709, higher than the trained model. The permutation result was an apparent indication that the features and the label were associated, but it could not tell us why. The label was built from each patient's change between baseline and follow-up, and building a label in this way can make it depend on the baseline values through measurement error and regression to the mean. That coupling could explain the model's performance without any forecastable biological response, and the trained model showed no improvement over the unfitted score.

This paper is about the distance between a check that passes and an interpretation that holds. We audited four of our own earlier analyses of public data, each of which had looked validated by the check we relied on, and found in all four that our interpretation of the score was unsupported. The technical details and the full simulation study are in a companion paper [1]. Here we report the four cases as the mistakes they were, because *how* a mistake survives a check is more useful to the next analyst than the fix itself. Our claim is narrow. We do not estimate how often such failures occur in the literature; we show that they can occur in careful work, and that the checks themselves give no sign that they have.

II. Literature Review

The three checks we relied on do different jobs. Each addresses a specific failure mode under stated assumptions; none establishes the full biological interpretation. Table I lists, for each check, the question it answers, the question it is easily read as answering, and the rival explanation it leaves open in our cases.

A. Cross-Validation

Cross-validation is not a hypothesis test; it estimates how well a model predicts samples it was not trained on, *collected the same way* as the training samples. Grouping folds by subject addresses one rival, that the model recognises individuals rather than classes. If every sample of one class was collected differently from the other, cross-validation will accurately report that the difference generalises.

B. Label Permutation Tests

A label permutation tests the null hypothesis that the features and the label are independent [2]. Under the test's assumptions, a small p-value is evidence that *some* association exists; it says nothing about where the association comes from. The permutations can be generated by shuffling the whole label vector before cross-validation [2], or, when labels are defined inside each fold, by shuffling each fold's training labels

TABLE I
THE QUESTION EACH CHECK ANSWERS, THE QUESTION IT IS OFTEN READ AS ANSWERING, AND THE RIVAL EXPLANATION IT LEAVES OPEN IN OUR CASES.

Check	Question it answers	Question it is read as answering	Rival explanation it leaves open	Case
Label permutation	Are features and label independent?	Does the model forecast a real individual response?	The change-based label depends on the baseline features through regression to the mean	1
ROAR	Does the model rely on the top-ranked features?	Are the top-ranked features biology?	The top features may reflect sample handling as well as, or instead of, physiology	2
Subject-grouped CV	How well does the model predict new subjects collected the same way?	Does the score reflect the disease?	Each class was collected under its own protocol	3
Subject-grouped CV	Same as above	Does the label mean what it is called?	The label encodes a different contrast	4

and keeping its held-out labels fixed. Calibration here means whether the test holds its rejection rate at the chosen significance level when the null hypothesis is true. The calibration of the resulting p-value depends on which scheme is used, and the scheme should be stated.

C. Feature Attribution and Remove-and-Retrain

SHAP and similar methods rank features by their contribution to a model's predictions. ROAR [3] removes the top-ranked features, retrains the model, and compares the drop in performance with that from removing random features. It asks whether the model *relies* on the top-ranked features. It does not ask whether those features are biology.

D. Regression to the Mean, Batch Effects and Leakage

The failure modes in our cases are not new. Regression to the mean has been described since at least 1962 [4]: measurement error and day-to-day variation make an extreme baseline move back towards the mean, so baseline appears to predict change even when nothing happens [5], [6], and a change score that shares a term with the baseline is mathematically coupled to it [7]. Batch and collection effects can be confounded with the biological variable of interest [8], [9], and pre-analytical handling can introduce exogenous compounds into samples [10], [11]. Leakage of many kinds has been catalogued across ML-based science [12], and reporting guidelines such as TRIPOD+AI ask authors to define the outcome and its relation to the predictors [13].

E. Differences Between This Study and Previous Research

Previous work describes these problems in general. What our cases add is that each passed a check commonly treated as sufficient, in work by analysts who knew of the problem in general and did not recognise it in their own data. We pair each failure with the check it passed and with a targeted control that exposes it, and we use a simulation to show that one common correction for regression to the mean is itself not sufficient.

III. METHODOLOGY

A. Process Overview

We audited four of our earlier analyses. For each, we restated the claim as it was originally made, identified the check it had passed, asked which rival explanation that check leaves open (Table I), and applied a control targeted at that rival. We kept the original models, because the subject of the audit is the checks, not the models. Table II summarises the four analyses.

B. Datasets

All three cohorts are public on MetaboLights [14]. MTBLS242 is serum ^{1}H-NMR from a bariatric-surgery cohort with 21 quantified metabolites [15], [16]; Cases 1, 2 and 4 use it. MTBLS1 is urine ^{1}H-NMR from people with type 2 diabetes and controls [17]; Case 3 uses it. MTBLS881 is serum ^{1}H-NMR from 47 cardiac-surgery patients, each sampled at two anatomical sites (left atrium, declared primary, and pulmonary artery) [18]; we used it only to repeat Case 1, with the design written down before any model was fitted. Cases 2 and 4 share one cohort and one classification task (preoperative vs. twelve-month samples), with different model specifications (Table II); they illustrate two interpretation failures in the same classification task, not independent replications.

C. Controls

1) Learn-nothing scorer: A fixed score built from each patient's baseline values and the expected direction of each marker, with no fitting. A trained model that does not beat it has shown nothing beyond what the coupling between outcome and baseline provides. We report the model–scorer gap on held-out predictions pooled within each cross-validation repeat, with a 95% interval from resampling subjects; because of the pooling, the gap is not exactly the difference of the mean per-fold AP values.

2) Pairing-destroyed null: We shuffle which follow-up sample belongs to which patient and recompute the outcome. This keeps the coupling but removes any real individual response. Shuffling also breaks the positive correlation between a patient's two visits, so the null has *stronger* coupling than

TABLE II
THE FOUR AUDITED ANALYSES. AP = AVERAGE PRECISION. CASE 1'S COHORT HAS ONE SAMPLE PER PATIENT, SO STRATIFIED CV IS ALSO GROUPED BY PATIENT. IN CASE 3, SUBJECT MEMBERSHIP FOR PATIENTS IS NOT DEPOSITED AND WAS INFERRED (SECTION IV-E).

Case	Data	Samples (subjects)	Model	Validation	Headline metric
1	MTBLS242, serum	71 (71)	Sign-constrained L2 logistic	20×5 stratified CV; per-fold training-label permutation with refit (1,000; calibration not verified)	PR-AUC, mean per-fold AP
2	MTBLS242, serum	177 (106)	Monotone XGBoost	10×5 patient-grouped CV; recursive ROAR (exploratory)	PR-AUC, mean of 50 fold APs
3	MTBLS1, urine	132 (40–42)	Monotone XGBoost	5-fold CV grouped by inferred subject; 20 seeds	PR-AUC, mean per-fold AP
4	MTBLS242, serum	177 (106)	XGBoost, unconstrained (original)	10×5 patient-grouped vs. naive CV (same task as Case 2)	ROC-AUC, mean over folds

the real data and can exceed the observed lift even when a real response exists. It is therefore read in one direction only: clearing it shows something beyond regression to the mean; failing to clear it shows only that nothing beyond it has been demonstrated.

3) Joint residualisation: Inside each training fold, each marker's change is regressed on all baselines and the residuals define the label. This addresses linear baseline–change coupling. It also changes what "responder" means, to improving more than the baselines predict, and it removes any true biological dependence of change on baseline together with the artefact.

4) Synthetic artefact feature: A feature equal to a multiple of the label plus Gaussian noise, which carries no biology, is passed through the same SHAP ranking and removal pipeline (10 draws per effect size).

5) Metadata and label-provenance audits: We cross-tabulate the label against every deposited metadata column, read the collection protocol for each class, and write down in one sentence what physically distinguishes the two classes.

D. Permutation Procedure in the Empirical Analyses

In every fold of the repeated cross-validation, we permuted the training labels, reran the penalty selection and the fit, and scored the unchanged held-out labels; the statistic was mean fold AP. This per-fold scheme was used in Case 1 and in its MTBLS881 repeat. We have not verified its calibration for these models, so we treat the resulting p-values as the output of an uncalibrated permutation diagnostic, not as independent proof of association or of successful validation. None of the conclusions below rests on their exact values.

E. Simulation

To see when label permutation is misled by a change-based outcome, we simulated cohorts with 11 outcome markers, whose persistent latent levels were correlated at 0.3 (so measured baselines correlate at $0.3r$), and ten auxiliary features. Each marker was measured at baseline and follow-up with reliability r, the share of variance that persists between visits, and the follow-up could include a forecastable individual response of strength γ; with $\gamma = 0$ no forecastable response exists. The outcome was "at least 6 of 11 markers change in the expected direction," and the model was ridge regression on the 21 baseline features, scored by average precision in 5-fold CV. We varied $n \in \{50, 100, 200, 500\}$, $r \in \{0.25, 0.5, 0.75\}$ and $\gamma \in \{0, 0.25, 0.5\}$, with 1,000 datasets per setting; $r = 0.25$ reproduces the baseline–change correlation of -0.62 observed in MTBLS242. Each dataset was tested four ways at $\alpha = 0.05$ with 200 null replicates, refitting the model on every replicate: label permutation of the raw-change outcome, the pairing-destroyed null, and per-marker and joint residualisation followed by permutation. The raw-change arm permuted the whole label vector before cross-validation. The residualised arms used fold-specific labels and shuffled the training labels within each fold, leaving held-out labels fixed. These simulations do not establish the calibration of our empirical procedure.

IV. EXPERIMENTAL AND RESULTS

A. Case 1: The Outcome That Predicted Itself

What we believed. A preoperative serum profile can tell which of 71 patients will improve on at least 9 of 11 markers twelve months after surgery.

Why the check passed. The baseline value x_0 of each marker was both a feature and a term in its change $\Delta = x_{12} - x_0$. In our data the median correlation between baseline and change was -0.62, so the baseline features and the change-based label were dependent, and the permutation diagnostic returned $p = 0.001$. That result did not show that the model had learned anything about who responds to surgery.

What we found. The learn-nothing scorer reached PR-AUC 0.709, above the model's 0.683. Under the pairing-destroyed null, scored with the unfitted scorer, the shuffled data gave a median lift of 2.04× over prevalence; refitting the full model on each shuffle gave 1.99×. Our model reached 1.57× (Fig. 1), so nothing beyond regression to the mean was demonstrated.

We then addressed linear baseline–change coupling by joint residualisation. Three changes came with this design, and they matter for reading the numbers. The responder rule became at least 6 of 11 markers, the three compounds of uncertain provenance from Case 2 were removed (18 features instead of

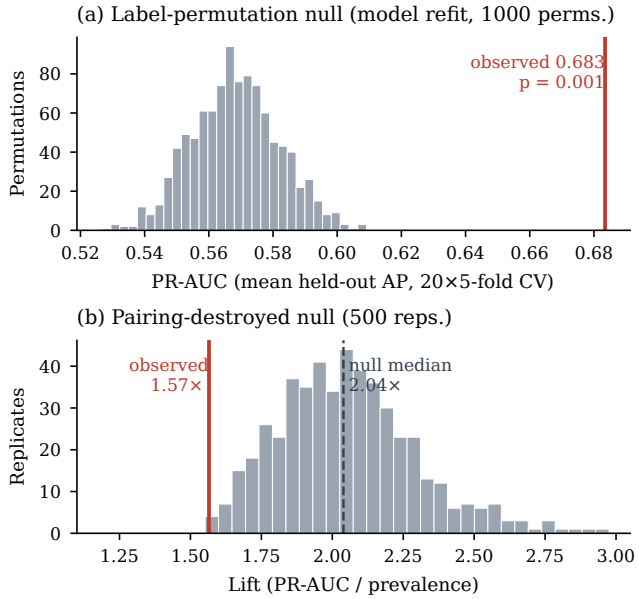


Fig. 1. Case 1, original design. (a) The historical permutation diagnostic (training labels permuted inside each fold, model refitted; calibration not verified), which returned $p = 0.001$. (b) Pairing-destroyed null of lift, computed with the unfitted scorer (median 2.04×; refitting the full model on each shuffle gives a median of 1.99×): the model does not clear it. This null is conservative, so failing to clear it means only that no response beyond regression to the mean has been shown, not that none exists.

21), and prevalence rose from 0.437 to 0.521, so PR-AUC levels are not comparable with the original design. Under this design the model reached PR-AUC 0.554 and the learn-nothing scorer 0.589 (means of per-fold AP). The pooled gap was -0.037, with a 95% subject-bootstrap interval of $[-0.111, +0.037]$. We found no evidence of improvement over the unfitted score; at $n = 71$, the interval cannot exclude a modest real gain.

B. Simulation: More Data, More Confident Rejections

Two simulation results surprised us more than the original error (Fig. 2). First, with no forecastable response and $r = 0.25$, raw-change label permutation rejected independence in 26% of datasets at $n = 50$ and in 100% at $n = 500$. Each rejection was correct about independence and wrong as evidence of a forecast: a larger sample estimates the coupling more precisely; it does not remove it. Second, per-marker residualisation, regressing each marker's change on its own baseline only, the correction applied in the corrected version of our analysis, still rejected in 8–29% of null datasets across settings, because correlated markers carry each other's baselines. Joint residualisation brought the rate to 5–8%, slightly above the nominal 5%. A correction is a check too, and it needs its own check.

C. Replication on a Second Cohort

In MTBLS881, on raw change scores the permutation diagnostic gave $p = 0.007$ at both sites. Neither site demonstrated an improvement over the learn-nothing scorer; the 95% interval for the model–scorer gap included zero at both sites (-0.037 $[-0.231, +0.118]$ at the left atrium and $+0.013$ $[-0.111, +0.154]$ at the pulmonary artery). The two sites come from the same patients, so they are correlated observations, not two replications, and their p-values come from the same uncalibrated per-fold procedure, here with an unconstrained model. Whether any real signal remains after joint residualisation, 47 patients could not tell.

D. Case 2: The Explanation That Was Faithful to the Wrong Thing

What we believed. A classifier separating preoperative from twelve-month samples (PR-AUC 0.973, CV grouped by patient) relied on metabolites that tell us how surgery changes metabolism, and ROAR confirmed which ones.

Why the check passed. Removing the SHAP-ranked features one at a time and retraining hurt the model more than removing random features, from the third removal on; these exploratory removal results suggest that the model relied on those features. But the first five removed included dimethyl sulfone, isopropanol and methanol. Isopropanol is used for skin preparation, and handling during collection can introduce exogenous compounds into samples [10], [11]; these sources describe the general risk, not contamination of this cohort. Dimethyl sulfone is a normal plasma metabolite that comes from diet, gut bacteria and the body's own sulfur metabolism [19], any of which could differ before and after surgery. We therefore treat the three as compounds of *uncertain provenance*: they may reflect biology, sample handling, or both, and an exogenous origin would not by itself imply contamination during handling. The deposited data cannot settle this, and the argument below does not depend on it.

What we found. Removing these three gave PR-AUC 0.954; removing three established metabolites gave 0.959. Both removals reduced performance below almost all random removals; the observed gap was 0.0056, against a fold-to-fold standard deviation of about 0.03, and we performed no formal equivalence test. ROAR, in other words, gave no sign of any difference in provenance between the two groups. With the synthetic artefact feature at an effect size similar to dimethyl sulfone's, SHAP ranked it first in 6 of 10 draws and second in the rest, and its median single-removal drop exceeded that of every real metabolite. One caveat applies: SHAP rankings were computed on all 177 samples, including those later used for evaluation, so the informed removals are subject to selection bias of unquantified magnitude [20]. We therefore treat the ROAR comparisons as exploratory; the synthetic feature supports the conceptual point without them.

The lesson. A faithfulness test checks whether the features an explanation ranks highly are the ones that drive the model's predictions; it does not certify the model as a whole. Removal-based faithfulness tests measure how much non-redundant signal a feature carries, whatever its source. It is faithful to the model, and the model is faithful to whatever separates the classes [21], [22]. ROAR cannot adjudicate provenance: it cannot tell us whether the top features reflect biology,

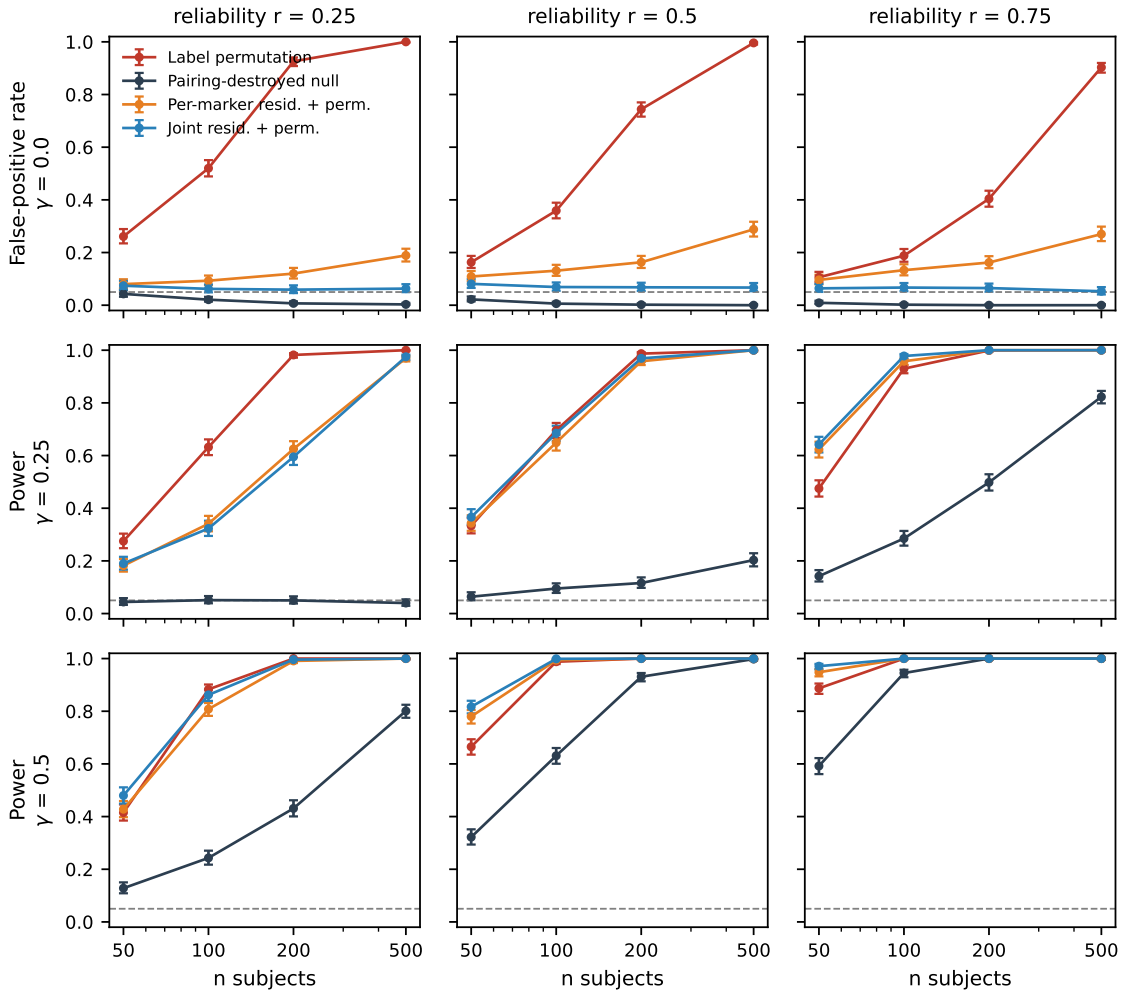


Fig. 2. Simulated rejection rate against the number of subjects, 1,000 datasets per point; error bars are 95% Wilson intervals for the rejection proportions. Rows vary the strength of forecastable response ($\gamma = 0$, 0.25 and 0.5); columns vary measurement reliability r. With $\gamma = 0$ there is no forecastable individual response, so every rejection in the top row (labelled false-positive rate) is a false signal of forecasting skill, even where it correctly rejects independence. On raw change, label permutation (whole-vector) rises towards 100% as n grows; per-marker residualisation also rises; joint residualisation stays at 5–8%. The lower rows show power when a forecastable response exists. Dashed line: $\alpha = 0.05$.

handling, or both, and for these three compounds the question stays open.

E. Case 3: The Cohort That Was Two Studies

What we believed. Urine NMR spectra separate people with type 2 diabetes from controls with PR-AUC 0.976, under CV grouped by subject.

Why the check passed. Grouping by subject stops the model from recognising a person it has already seen. It cannot detect a difference shared by every member of a class.

What we found. The 0.976 came from one random seed; over 20 seeds the mean was 0.911 (range 0.845–0.975). We then cross-tabulated the label against all 64 columns of the deposited metadata. Fifty-six were constant, and apart from per-sample identifiers only sex and the label varied; no column recorded a date, batch or run order. The deposited protocol, however, described two different studies: 12 volunteers aged 18–55 sampled at seven fixed times, and patients aged 30–65 with higher body-mass index sampled one to three times. Every control sample name began with `ADG19007u` and every patient sample name with `ADG10003u`. Cohort membership, and with it the sampling schedule and any difference in how the two studies collected and stored urine, is therefore perfectly confounded with the label. Age and body-mass index were recruited under different but overlapping ranges, so they are potential confounders as well; neither is recorded per sample.

One methodological limit of our own audit belongs here. Which samples come from which patient is not deposited, so we grouped patient samples with a numbering heuristic. It merged at least three pairs of distinct patients, which is conservative for those pairs, but we cannot rule out that it also split one patient's samples across groups. Grouping does change the score: over the same 20 seeds, grouped CV gave 0.911 (0.906 under an alternative, protocol-consistent

grouping) and ungrouped CV 0.969. The protocol problem, however, does not depend on grouping: it is a property of the study design, and no choice of folds can remove it.

The lesson. The classifier may have learned diabetes, collection protocol, or both; these data cannot tell which, because each protocol contains only one class [8]. Because these data cannot separate disease from collection protocol, whether the classifier's performance reflects a difference in disease remains undetermined.

F. Case 4: The Label That Was Named for Something Else

What we believed. Serum NMR separates obese from non-obese people with ROC-AUC 0.973, the AUC of the mean ROC curve over a single stratified 5-fold split (0.974 when we re-ran it).

Why the check passed. When we repeated the evaluation 10 times with the original model, CV grouped by patient and ungrouped CV both gave a mean fold ROC-AUC of 0.965 (a different summary from the mean-curve AUC; the re-run split's mean fold AUC was 0.976), although in ungrouped CV 65% of test samples had the same patient's other visit in the training folds. Removing that leakage did not lower the score, and the agreement looked reassuring.

What we found. The "non-obese" class was the twelve-month post-surgery visit of the same cohort. Of the 106 patients, 71 appear in both classes and supply 80% of the samples. The metadata record no body-mass index, so we cannot check whether any twelve-month patient was non-obese. The score reflects whatever differs between a person before bariatric surgery and the same person a year later: weight loss, but potentially also diet, supplements and medication after surgery, and a year's difference in sample storage. These data cannot attribute it to obesity.

The lesson. The correct name is "pre- vs. post-surgery classification."

G. Common Pattern Across the Four Cases

In each case, the check indicated that the data contained predictive structure, with the caveats already noted for our permutation procedure and the exploratory ROAR comparisons. What it could not establish was where that structure came from: from the arithmetic of the outcome, from compounds of uncertain origin, from the collection protocol, or from surgery. Each of these rivals would pass the check as easily as the biology we had in mind, so passing could not distinguish them. Three habits made this easy to miss.

We reported checks, not rivals. A methods section lists the checks that were run. It rarely lists the alternative explanations those checks were meant to address. Once a check is listed, it reads as a general certificate of validity, though it addresses only one failure mode.

We trusted bigger numbers more. A smaller p-value, a larger n, or an AUC near 1 feels like stronger evidence. Case 1 and the simulation show a setting in which more data produced a more confident but equally uninformative rejection, and Case 3 one in which a high, seed-dependent score sat on top of a design confound. In small cohorts, a score close to 1 is a reason to look for a proxy, not a reason to stop.

We checked the model, not the label. All our checks took the label as given. Two of our four failures were in the label itself (Cases 1 and 4), and a third (Case 3) was in how its classes were collected. No amount of model validation reaches a problem that sits upstream of the model.

V. Conclusion

Each of our four analyses produced a result we treated as validated, and in each the biological interpretation we attached to it was unsupported or unidentifiable. The responder forecast shows no evidence of improvement over an unfitted score at $n = 71$; this is absence of evidence, not evidence that no forecast is possible. The exploratory ROAR results suggest which features the model relies on, not where those features come from. For the diabetes classifier, these data cannot show whether its performance reflects disease or collection protocol. The obesity classifier is a pre- vs. post-surgery classifier. A check that passes addresses one failure mode under its own assumptions; it does not establish that the interpretation is right.

These conclusions have limits. We audited our own analyses of three small public datasets, chosen because their problems can be verified from the published files, and we make no claim about how common these failures are in the literature. Our findings concern validation practice and say nothing for or against the biology of obesity, bariatric surgery or diabetes. We have not shown that any classifier used *only* an artefact; we have shown that the checks could not rule the artefact out. The provenance of the three compounds in Case 2 is not established. The permutation p-values in Case 1 and its replication come from a per-fold procedure whose calibration we have not verified. The simulation uses Gaussian data, a linear model and one outcome rule, and the joint residualisation it supports also removes any true dependence of change on baseline. The checks we discuss can be useful when matched to their assumptions and target; our point is that passing them is not sufficient for a biological interpretation.

VI. Future Directions

For future small-n metabolomics studies, including our own, we recommend naming the rival explanations for a headline score before choosing the tests, and asking six questions; the companion paper gives reference code for each [1].

1) *Is the outcome computed from the features?* If so, a label permutation is not evidence of a forecast. If the scientific target is change beyond what baseline predicts, regress change on *all* baselines inside each training fold; if baseline-dependent change is itself of interest, this correction removes it, and the question may require repeated baseline measurements or another design that separates measurement error from response. In either case, report a pairing-destroyed null as a one-sided diagnostic.

2) *Was the model refitted inside every permutation, and is the permutation scheme stated?* If the observed statistic sits below the whole null, audit the null construction, the scoring pipeline, the model constraints and the fitting procedure before reading the p-value.
3) *Does the model beat a score that learns nothing?* Report the gap with an interval that resamples subjects.
4) *Could any feature come from sample handling or drugs?* Report results with and without such features, and run a feature known to carry no biology through the attribution pipeline.
5) *Does any metadata column or collection protocol line up with the label?* Cross-tabulate the label against every column and read how each class was collected.
6) *What, in one sentence, physically distinguishes the two classes?* If the sentence differs from the title, change the title.

These questions can be asked of the existing records, although resolving some of the problems they reveal requires new measurements or a different study design. For the present audit, four analyses would strengthen the evidence: rerunning the Case 1 raw-change test with whole-vector permutations, to assess how sensitive the conclusion is to the permutation scheme; estimating the rejection rate of the per-fold scheme on repeated simulated data under the null hypothesis it is meant to test, to check its calibration (keeping "no forecastable response" distinct from "features and label are independent," which are not the same condition); computing the Case 2 SHAP rankings inside each training fold; and repeating Case 4 in a cohort with recorded body-mass index and lean controls.

Data and Code Availability

MTBLS242, MTBLS1 and MTBLS881 are public through MetaboLights [14]. The analysis code, the simulation and a record of every reported number with its script and random seed are available from the authors.

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